Exponential Equations

Exam code: 9MA0

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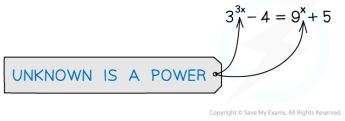
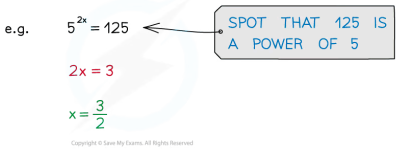
Exponential equations

What are exponential equations?



- An equation where the unknown is a power
- In simple cases the solutions can be "spotted"



How do I solve exponential equations?

- Laws of Indices are often needed to rewrite equations
- Laws of Logarithms are used to solve exponential equations
- **ln** (**$\log_e$**) is often used
- Answers are often written in terms of **ln**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

ARE INTEGERS TO BE FOUND

STEP 1: TAKE LOGARITHMS OF BOTH SIDES

$\ln 3^{2x-1} = \ln 2^{6-x}$

STEP 2: USE LAWS OF LOGARITHMS TO REMOVE POWERS

$(2x-1)\ln 3 = (6-x)\ln 2$

STEP 3: REARRANGE TO ISOLATE x

$2x\ln 3 - \ln 3 = 6\ln 2 - x\ln 2$

$2x\ln 3 + x\ln 2 = 6\ln 2 + \ln 3$

$x(2\ln 3 + \ln 2) = 6\ln 2 + \ln 3$

$x = \dfrac{6\ln 2 + \ln 3}{2\ln 3 + \ln 2}$

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STEP 4: USE LAWS OF LOGARITHMS TO REWRITE IN CORRECT FORM

$x = \dfrac{\ln 2^6 + \ln 3}{\ln 3^2 + \ln 2}$

$x = \dfrac{\ln 64 + \ln 3}{\ln 9 + \ln 2}$

$x = \dfrac{\ln 192}{\ln 18}$ ← $p = 192$, $q = 18$

$\log_a x + \log_a y = \log_a xy$

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e.g. SOLVE $6^{3x+2} = 42$, GIVING YOUR ANSWER CORRECT TO 3 SIGNIFICANT FIGURES

$6^{3x+2} = 42$

$\ln(6^{3x+2}) = \ln 42$ — TAKE LOGARITHMS OF BOTH SIDES

$(3x+2)\ln 6 = \ln 42$

$3x+2 = \dfrac{\ln 42}{\ln 6}$ ← $\neq \ln 7$

$3x = \dfrac{\ln 42}{\ln 6} - 2$

$x = \dfrac{\dfrac{\ln 42}{\ln 6} - 2}{3}$

$x = 0.0286777...$ ← AVOID USING CALCULATOR UNTIL NECESSARY

$x = 0.0287$ (3 sf)

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How do I solve hidden quadratic exponential equations?

- Look out for "hidden" squared terms, these are hidden quadratics which will need to be solved
  - $4^x = (2^2)^x = 2^{2x} = (2^x)^2$
  - $e^{2x} = (e^2)^x = (e^x)^2$

"HIDDEN" SQUARED TERM

$21e^x - 4 = 5e^{2x}$

$5(e^x)^2 - 21(e^x) + 4 = 0$

REARRANGE TO QUADRATIC FORM

$(5e^x - 1)(e^x - 4) = 0$

USE $y = e^x$ IF PREFERRED

$e^x = \frac{1}{5}$ $\quad$ $e^x = 4$

$x = \ln\frac{1}{5}$ $\quad$ $x = \ln 4$

$x = \ln 5^{-1}$

REWRITE IN CORRECT FORM

$x = -\ln 5$ $\quad$ $x = \ln 4$

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